# Soloman Northrop

Lead Software Engineer | Full-Stack, Robotics and AI Platforms | C#/.NET, TypeScript, Flutter, Unity

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## SUMMARY

Lead software engineer with seven years of production work. I build complete platforms as the only engineer on them. That includes a Flutter robotics control platform running on a commercial security robot, a WebXR physical-therapy platform used in 15+ clinics, and a multi-tenant cloud platform for streaming and AR/VR. My main languages are C#/.NET and TypeScript. I also ship in Flutter, Unity and Python. I have put LLMs, on-device computer vision and physics simulation into products that customers use. I own architecture through release and work directly with founders, clinicians and hardware engineers.

## SKILLS

**Languages:** C#, TypeScript/JavaScript, Dart, Python, SQL

**Backend and data:** ASP.NET Core, Entity Framework, Node.js, Express, Colyseus, PostgreSQL, SQL Server, Supabase, REST APIs, webhooks, multi-tenant architecture, Hangfire

**Front end and apps:** React, Vite, Svelte, Three.js, Flutter, Unity, WebXR, Expo/React Native, Electron, Razor

**Cloud and delivery:** AWS (ECS Fargate, ALB, EFS, ECR, SES), Fly.io, Cloudflare, Auth0, Docker, GitHub Actions, CI with automated tests (node:test, Playwright, flutter_test), multi-platform release builds

**AI and robotics:** LLM integration (OpenAI including realtime voice, Gemini, Claude), agent orchestration and MCP, on-device inference (ONNX Runtime, TFLite, TensorRT on Jetson), YOLO, depth and segmentation deployment, ZED stereo vision, GNSS/RTK, mapping and navigation, physics simulation (Rapier, WebGPU), serial and firmware protocols

## EXPERIENCE

**Lead Software Engineer (full-time)** | Robocor — Apr 2024 – Present

*Remote, Florida*

- Designed and built Robocor 2.0, a Flutter/Dart robotics control platform for iOS, Android, macOS, Windows, Linux and web. Release pipelines cover Windows, macOS and Jetson/Linux. It handles autonomous navigation, telemetry, computer vision, fleet management and remote operation. The backend is Supabase with multi-tenant worker services on Fly.io. Sole software engineer, reporting to the founder and working with the hardware and firmware team.
- Shipped the Guardian security robot to a live customer site (Nimnicht Chevrolet). Wrote the whole software stack, from the firmware serial protocol and motor watchdog up to the cloud backend and operator app.
- Built the on-device perception pipeline on NVIDIA Jetson with a ZED stereo camera. YOLO26 and YOLO-World for detection, Depth Anything v2 and UniDepth for metric depth, SegFormer for segmentation. Models are exported to ONNX Runtime, TFLite and TensorRT, through a Flutter ONNX Runtime plugin I maintain. Fine-tuned a custom segmentation model as a field experiment.
- Added GNSS/RTK path following, offline mapping and navigation, and an OpenAI GPT-4o operator agent with realtime voice so staff can talk to the robot.
- Created corBench, a three.js/WebGPU and Rapier physics test bench. Virtual sensor boards drive the real parsers unchanged. 580+ recorded runs. Eight saved docking scenarios run as regression tests from 27 start poses each; 212 of 216 starts dock.

**Software Developer III, Full Stack** | VEERA Medical — Apr 2023 – Present

*Hybrid, Florida*

- Built and launched MedXR (Recovr360), a white-label WebXR physical-therapy platform. The patient wears a Meta Quest headset; the practitioner works from a browser. In use at more than 15 clinics for home and in-clinic therapy. Live at app.recovr360.com.
- Wrote all of it. The Node/TypeScript real-time backend uses Colyseus rooms with a strict session lifecycle and Express APIs. The client is Svelte and Three.js WebXR, plus a Unity scene exporter and a native Quest client. Also the patient progress pipeline and clinic monitoring.

- Migrated the platform from Fly.io to AWS ECS Fargate behind an ALB, with EFS, ECR and Cloudflare. Rolled out Auth0 PKCE login behind feature flags.
- Added accessible patient input: Xbox Adaptive Controller, wheelchair joystick HID adapters and standard gamepads.
- Presented the platform live at ACRM 2023. A practicing clinician ran the wireless demo.

**Software Developer II** | VEERA Medical Oct 2021 – Mar 2023

- Built the first VEERA Medical cross-platform application in C#, Unity, JavaScript and ASP.NET Core, on the Reach Cloud codebase I co-authored. Razor UIs, authentication and the clinician data pipeline.

**Co-Founder and Lead Engineer (side venture)** | Reach Cloud Jan 2022 – Present

*Remote, St. Petersburg, FL*

- Built and maintain a multi-tenant cloud platform for streaming, remote device management and cross-platform multiplayer VR/AR/MR, with a Unity SDK and public API. Live at reachcloud.org.
- Wrote the developer docs and the C# generator that builds them. Set the engineering roadmap and shipped across platforms.

**Junior Software Application Programmer (part-time)** | Cardno May 2019 – Sep 2023

*Clearwater, FL*

- Maintained and extended the Georgia DOT permit system in C# and SQL. Added batch reporting and export, and Hangfire background processing. Made the site faster.

## INDEPENDENT PRODUCTS

- Solla Code (t3.codes): open-source fork of T3 Code that I maintain and extended into a control surface for AI coding agents. Multi-agent orchestration, MCP tools, scheduled autonomous tasks, shared browser automation, approval gating and usage guards. 850+ commits of my own since July 2026 across the Electron, web and Expo mobile clients. Nightly prereleases and tagged stable releases.
- SkyCal (skycalapp.com): family calendar and planning SaaS in React, TypeScript and Vite on Supabase. Stripe household billing, Google and Outlook calendar sync (Microsoft Graph), a GPT realtime voice assistant and a referral program; 50+ test files including Stripe sandbox tests.
- Glow Music Therapy platform (glowmt.com): public site, family portal and practice-management backend in framework-free Node and PostgreSQL. WebAuthn passkeys, RBAC, an append-only audit trail, Stripe invoicing and HIPAA-aligned deployment gates. In production for a paying practice.
- GenEdit Pro: open-source browser-first AI video editor on ffmpeg.wasm (React, Zustand, Playwright tests).

## EDUCATION

**St. Petersburg College** | Associate of Science, Computer Programming | 2016 – 2017